

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I Thomish (191412024) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

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Signature of the student:

(89)

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Solve the 6-Queens Problem: place 6 queens on a 6x6 chessboard such that no two queens attack each other. Formulate the Problem components (state space, initial state, succession function, goal test) and explain how incremental formulation reduces the search space compared to complete-state formulation.

Solution

Problem Formulation:

1) State Space:

The state space consists of all possible configurations of placing 0 to 6 queens on a 6x6 chessboard, ensuring that each queen occupies a different row.

2) Initial state:

- * Empty 6x6 chessboard
- * No queens are placed.

3) Succession Function:

- * Place ~~one~~ queen in the next row.

- * Choose only columns where the queen is not attacked by

Previously placed queens.

- * Continue until all six queens are placed.

d) Goal Test:

The goal is reached when:

* Exactly 6 queens are placed.

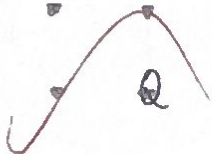
* No two queens attack each other (same row, same column, or diagonal).

One valid solution

Row	column
1	1
2	4
3	6
4	1
5	3
6	5

Block Representation:

.	Q	.	.	.	.
.	.	.	Q	.	.
.	.	.	.	.	Q
Q	.	.	.	.	.
.	.	Q	.	.	.
.	.	.	.	Q	.



(Queen = Q)

Generalization in social life

Generalization

- there is a group of people
- social norms are being changed
- number of generalizations is very large
- people will still be general

Generalization

- there are two types of people
- immediately right before every other
- they will still be general in general
- people will still be general

Form

Generalization	Generalization
there is a group of people	there are two types of people
they will still be general	they will still be general
they will still be general	they will still be general
they will still be general	they will still be general

2/ For a chess-playing agent, characterize the task environment across all PEAS/environment dimensions (observable, deterministic, episodic, static, discrete, sequential). Justify each classification with reference to the rules of chess.

PEAS:

PEAS

Performance Measure:

- * Win the game
- * Checkmate the opponent
- * Minimize mistakes.
- * Maximize strategic advantage

Environment:

- * Chessboard
- * Pieces
- * opponent

Actuators:

- * Move chess pieces

Sensors:

- * observe board position
- * Detect legal moves.
- * Detect opponent moves.

Environment characteristics

1. Observable

Fully observable

Reason:

- * Both players can see the complete chessboard
- * No hidden information exists

2. Deterministic

Deterministic

Reasons:

- * Every move has a predictable outcome
- * No randomness or chance is involved

3. Episodic or Sequential

Sequential

Reason:

- * Every move affects future moves.
- * Current decisions influence the entire game.

4. Static or Dynamic

Static

Reason:

- * The board does not change while player is thinking
- * Time does not alter the board state.

5. Discrete on continuous

Discrete

Reason:

- * Finite number of legal moves.
- * Board has fixed squares.

6. Single-Agent on Multi-Agent

Multi Agent

Reason:

- * Two intelligent players compete
- * one player's actions affect the other.

Summary Table

Property	Classification	Reason
observable	Fully observable	Entire board visible
Deterministic	Deterministic	No randomness
Episodic	Sequential	Moves affects future
static	static	Board unchanged while thinking
Discrete	Discrete	Two competing players
Agents	Multi-Agent	Finite legal moves

An exploration robot is sent into an unknown cave system with no prior map, & its sensors have limited range. Explain how the robot can use search with Partial information (contingency / online search) to build up a map while searching for exit, & discuss the risks of irreversible actions in this environment.

solution:

Problem:

The robot.

- * Has no map.

- * sensors have limited range

- * Must search for the exit while building a map.

online (contingency) search

Since the robot has incomplete information, it performs online search.

step 1:

start from the entrance

step 2: sense nearby passages using sensors

step 3: update an internal map with discovered locations

step 4: choose an unexplored path

step 5: Repeat until the exit is found.

As exploration continues, the map becomes more complete

Building the Map:

The robot records:

- * Visited locations
- * unexplored paths
- * obstacles
- * Dead ends
- * This gradually creates a complete map of the cave

Contingency Search:

Because information is uncertain:

- * The robot continuously scans the environment.
- * It changes its plan based on newly discovered paths
- * It can backtrack if necessary.

Risks of Irreversible Actions

Some actions cannot be undone.

Examples:

- * Falling into a deep pit
- * Entering a collapsing tunnel.
- * Crossing a one-way Passage
- * Running out of battery after taking a wrong route

These may prevent the robot from reaching the exit.

How to Reduce Risks

- * Explore cautiously
- * Avoid unknown dangerous path
- * Keep track of visited locations
- * Reverse battery power.
- * Backtrack whenever possible before entering risky areas

Conclusion:

An exploration robot uses online (contingency) search to explore an unknown cave, gradually builds a map from sensor data, and continuously its decisions based on new information, careful exploration & avoiding irreversible actions help improve the chances of safely finding the exit.

